

Discovery Document

Food & Grocery Delivery Analytics Platform

1. Project Overview

Project Name

Food & Grocery Delivery Analytics Platform

Project Type

Cloud-Based Analytics & Business Intelligence Solution

Business Domain

Food Delivery / Grocery Delivery / Retail Analytics

Reference Business Model

The project simulates analytics operations similar to DoorDash Grocery delivery operations in Canada.

Objective

Develop a centralized analytics platform to analyze restaurant delivery operations and grocery commerce insights using cloud analytics technologies.

Proposed Solution

The solution includes:

- Google Sheets for staging and lightweight data cleaning
- BigQuery for cloud-based data warehousing
- Google Data Studio for dashboard visualization
- SQL analytics for operational reporting

- Agile delivery methodology for project execution
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2. Business Problem Statement

The organization currently lacks centralized visibility into food and grocery delivery operations across multiple Canadian cities.

Current reporting processes are fragmented and rely heavily on manual spreadsheets, making it difficult to:

- Monitor delivery performance
- Analyze restaurant and grocery trends
- Compare city-level operations
- Track customer satisfaction metrics
- Identify high-performing categories and stores
- Enable scalable cloud reporting

The absence of a centralized analytics platform results in delayed reporting, inconsistent insights, and limited operational visibility.

3. Discovery Deliverables

Deliverables Included

Deliverable	Description
Stakeholder Register	Stakeholder identification and impact analysis
User Personas	Target user profiles and goals
Current State Documentation	Existing process analysis and pain points
Requirements Documentation	Themes, epics, and user stories
Project Roadmap	Phased implementation plan
Success Metrics	KPIs and acceptance criteria
Architecture Design	Cloud analytics architecture
Process Flow Diagrams	As-Is and To-Be flows

4. Stakeholder Register

Stakeholder Analysis

Stakeholder	Role	Responsibility	Influence	Interest
Product Owner	Business Sponsor	Defines business vision	High	High
Business Analyst	Requirements Lead	Gathers and documents requirements	High	High
Data Analyst	Reporting & Insights	Creates dashboards and KPIs	Medium	High
Operations Team	Delivery Monitoring	Reviews operational performance	Medium	High
Engineering Team	Data Pipeline Support	Maintains integrations and warehouse	Medium	Medium
Marketing Team	Campaign Analytics	Uses customer and category insights	Low	Medium
Business Leadership	Strategic Decisions	Reviews executive reports	High	Medium
Customers	End Users	Place restaurant and grocery orders	Low	High

5. User Personas

Persona 1 — Operations Manager

Name

Michael Thompson

Role

Operations Manager

Goals

- Monitor delivery performance

- Reduce delays
- Improve operational efficiency
- Compare city-level delivery trends

Pain Points

- Manual reporting delays
- Lack of real-time visibility
- Difficulty identifying low-performing locations

Analytics Needs

- Delivery time dashboards
 - City-level comparisons
 - Rating trends
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Persona 2 — Business Analyst

Name

Sophia Patel

Role

Business Analyst

Goals

- Create KPI dashboards
- Analyze customer behavior
- Support business decisions

Pain Points

- Data inconsistencies
- Multiple disconnected files
- Slow data preparation

Analytics Needs

- Centralized warehouse
 - Interactive dashboards
 - Self-service reporting
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Persona 3 — Product Owner

Name

Daniel Chen

Role

Product Owner

Goals

- Expand grocery analytics capabilities
- Increase customer retention
- Improve business visibility

Pain Points

- Limited operational transparency
- Lack of scalable reporting

Analytics Needs

- Executive dashboards
 - Revenue trends
 - Strategic KPIs
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6. Current State Documentation

Existing Process (As-Is)

Current Workflow

Restaurant and grocery data are collected from multiple sources and manually analyzed using spreadsheets.

As-Is Process Flow

1. Restaurant data collected manually
2. Grocery data collected separately
3. Data stored in spreadsheets
4. Manual cleaning and formatting
5. Excel-based analysis

6. Static reports shared with stakeholders

Current Challenges

Problem Area	Description
Fragmented Data	Data stored across multiple files
Manual Reporting	Significant manual effort required
Delayed Insights	Slow report generation
Data Quality Issues	Missing and inconsistent values
Limited Scalability	Difficult to support increasing data volume
Reporting Limitations	No centralized cloud analytics

7. Future State Documentation

Proposed Process (To-Be)

To-Be Process Flow

1. Data collected through structured sources
 2. Cleaning performed in Google Sheets
 3. Federated access enabled in BigQuery
 4. Native BigQuery tables created for analytics
 5. SQL views built for reporting
 6. Interactive dashboards created in Google Data Studio
 7. Stakeholders access centralized insights
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8. Functional Requirements

Requirement Themes

Theme ID	Theme	Description
TH-1	Restaurant Analytics	Analyze restaurant delivery operations
TH-2	Grocery Analytics	Analyze grocery products and stores

TH-3	Cloud Reporting	Centralized analytics platform
TH-4	Dashboarding	KPI visualization and filtering
TH-5	Operational Insights	Delivery and ratings analysis

9. Epics

Epic ID	Epic Name	Description
EP-1	Restaurant Reporting	Restaurant delivery KPI analysis
EP-2	Grocery Reporting	Grocery category and pricing analytics
EP-3	Dashboard Development	Create cloud dashboards
EP-4	Data Integration	BigQuery integration and SQL views
EP-5	User Filtering	Interactive dashboard filtering

10. User Stories

Story 1

User Story

As an operations manager,
I want to analyze delivery times by city,
So that I can identify operational delays.

Acceptance Criteria

- Dashboard displays delivery time trends
 - City-level filtering is available
 - Metrics refresh correctly
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Story 2

User Story

As a business analyst,
I want to compare restaurant ratings,
So that I can identify high-performing vendors.

Acceptance Criteria

- Dashboard displays average ratings
 - Ratings can be filtered by city
 - Visuals update dynamically
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Story 3

User Story

As a product owner,
I want grocery category analytics,
So that I can understand customer demand.

Acceptance Criteria

- Grocery categories displayed visually
 - Product distribution available
 - Filters function properly
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Story 4

User Story

As a dashboard user,
I want location-based filters,
So that I can analyze regional performance.

Acceptance Criteria

- Filters work across all visuals
 - Dashboard performance remains stable
 - Correct records displayed
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11. Non-Functional Requirements

Requirement Type	Description
Performance	Dashboard should load within 5 seconds
Scalability	BigQuery should support increasing datasets
Availability	Dashboards available to stakeholders
Security	Access controlled through GCP IAM
Usability	Dashboards should be easy to navigate
Reliability	Data refresh should remain consistent

12. Data Sources

Source	Purpose
DoorDash Canada Dataset	Restaurant analytics
Grocery Dataset	Grocery analytics
Google Sheets	Staging and cleaning
BigQuery	Centralized cloud warehouse

13. Solution Architecture

Proposed Architecture

Google Sheets
↓
BigQuery Federated Tables
↓
Native BigQuery Tables
↓
SQL Views
↓
Google Data Studio Dashboards

↓
Business Users

14. Project Roadmap

Phase 1 — Discovery & Planning

Activities

- Stakeholder analysis
- Requirement gathering
- Discovery workshops
- Architecture planning

Deliverables

- Discovery document
 - Stakeholder register
 - Initial roadmap
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Phase 2 — Data Preparation

Activities

- Data cleaning
- Google Sheets preparation
- Data validation

Deliverables

- Clean datasets
 - Validation rules
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Phase 3 — Cloud Integration

Activities

- BigQuery setup

- Federated access configuration
- SQL view creation

Deliverables

- Cloud warehouse
 - Analytics views
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Phase 4 — Dashboard Development

Activities

- KPI design
- Dashboard creation
- Visualization testing

Deliverables

- Operational dashboards
 - Interactive reporting
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Phase 5 — Testing & UAT

Activities

- User acceptance testing
- Dashboard validation
- Performance optimization

Deliverables

- UAT sign-off
 - Final dashboards
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15. Agile Delivery Approach

Methodology

Sprint Structure

Sprint	Focus Area
Sprint 1	Discovery & Cleaning
Sprint 2	BigQuery Integration
Sprint 3	Dashboard Development
Sprint 4	Testing & Optimization

Agile Ceremonies

Ceremony	Purpose
Sprint Planning	Define sprint scope
Daily Standup	Track progress
Sprint Review	Demonstrate work
Retrospective	Identify improvements

16. Risks & Mitigation

Risk	Impact	Mitigation
Data Quality Issues	Incorrect reporting	Validation rules
Permission Errors	Dashboard failures	IAM access management
Performance Delays	Slow dashboards	Native BigQuery tables
Inconsistent Data	KPI inaccuracies	Cleaning standards

17. Success Metrics

Business KPIs

KPI	Target
Dashboard Load Time	< 5 seconds
Data Accuracy	> 95%
Reporting Efficiency	Improved reporting cycle
Delivery Analytics Coverage	Multi-city visibility
Grocery Category Insights	Complete category reporting

Acceptance Criteria

Area	Acceptance Criteria
Dashboards	Interactive filters working
Data Accuracy	Clean and validated data
Reporting	KPIs display correctly
BigQuery	Queries execute successfully
Visualization	Charts refresh dynamically

18. Future Enhancements

Planned Enhancements

- Real-time streaming analytics
 - Predictive delivery forecasting
 - Customer segmentation
 - Machine learning recommendations
 - Advanced operational KPIs
 - Mobile dashboard optimization
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19. Conclusion

The proposed Food & Grocery Delivery Analytics Platform provides a scalable cloud-based analytics solution that centralizes restaurant and grocery reporting operations.

The solution enables stakeholders to monitor operational KPIs, delivery performance, customer ratings, and grocery analytics using BigQuery and Google Data Studio.

The project follows Agile delivery methodology and supports scalable cloud analytics for future business expansion.